

Corpus & Engine

Formalization and
broad-scale applications

A source-faithful account, an explicit mathematical
specification and a reproducible research package.

THE GENERATIVE THREAD

Grammar / Geometry / Time / Topology
 $0 \longrightarrow 1 \quad 1 \longrightarrow 01 \quad \rho = \log_2(f/f_{\text{ref}}) \quad d = 0$

Prepared for Tom Klootwijk

Identifier as supplied NL200678942

Date as supplied 10-07-1990

Issued 10 September 2026

Research dossier, not a certificate of identity, ownership or physical validation.

Original corpus exports and newly introduced engineering choices are distinguished.

Contains personal identifying details supplied by the requester.

Executive brief

The central proposition is a shared generative recipe for instructions, multiscale coordinates, geometric boundaries and time-dependent output. In the final stage of the supplied corpus, binary grammar execution replaces the earlier golden-ratio search tree. The intended engine is therefore not simply a synthesizer with an unusual waveform: it is a proposed way to organize and generate structured states before choosing how those states are visualized, sonified, simulated or encoded. [S1, pp. 19–22; S2, lines 83–99]

This edition gives that proposition three distinct forms. First, it records the **corpus as stated**, retaining its five-layer organization and characteristic terminology. Second, it supplies an **explicit engineering formalization**, including definitions that the source leaves open. Third, it provides a **bounded software demonstrator and application programme**, with reproducible checks rather than assumed physical performance.

Working formal definition

The engine is a grammar-addressed procedural framework in which finite binary words are interpreted as state transitions; those states address phase, scale, geometric fields and optional topological charts; typed output adapters then generate data or signals. A physical application additionally requires a model of the medium, devices, energy exchange and measurement.

This definition is the editorial engineering formulation adopted in this edition, not a verbatim sentence from the supplied corpus.

The strongest initial application candidates are procedural sound, synchronized audiovisual content, educational models, compact deterministic assets and bounded embedded-control sequences. Geometry, sensing, tomography and inverse-design applications are credible **research directions for selected components**, not established capabilities of the complete engine. The application portfolio specifies a baseline and a validation question for every proposed use.

The accompanying reference code reproduces the stated productions and blend formula, implements a declared interpretation and quotient chart, and separates grammar bits from a one-bit signal quantizer. **Thirty-five automated tests pass in the recorded environment.** This establishes the listed software properties only. No acoustic beam, physical energy recirculation, hardware efficiency gain, general absence of aliasing or universal golden-ratio optimality has been demonstrated.

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Edition: 1.0 · **Issued:** 10 September 2026 · **Reference code:** 0.1.0.

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1. Identity, scope and evidential method

1.1 What “corpus” means here

The corpus is the **body of text supplied by the requester**, not a medical record, an administrative dossier or an independently recovered collection of all earlier work. This edition covers two artifacts: the 23-page PDF `Tone.pdf` and the markdown export `Pasted markdown(9).md`. The latter explicitly clarifies that “corpus” refers to the body of text. Nothing outside these attachments is treated as an undisclosed part of the author’s original system. [S2, lines 131–133]

S1 — `Tone.pdf`. This is the longer conceptual progression. It begins with tone, musical notes, controller mappings and lookup tables; moves through procedural fields, log-polar textures, acoustic and caustic analogies, blend operators and spacetime; then introduces a golden-ratio tree, a Klein-bottle construction and finally a binary L-system. The closing grammar stage is important because it supersedes rather than merely supplements the previous tree. [S1, pp. 2–8, 13–22]

S2 — `Pasted markdown(9).md`. This is a later organized restatement and application discussion. Its central section supplies five Roman-numbered layers, from one-bit data through geometry, topology and grammar to output. Those five layers determine the organization of the source-faithful account below. [S2, lines 48–99]

Both artifacts contain exported AI/search dialogue, including generated explanations and confidence language. The requester attributes the corpus and its organizing concept to himself. That attribution is recorded, while the authorship of every exported sentence and every interlocutor’s contribution is not independently established. A conversation’s statement that an engine is “perfect” is treated as a statement in the corpus, not as experimental evidence.

1.2 Two records, not a silent rewrite

The **authorial record** preserves what the supplied documents say, including metaphors, ambiguities and changes of direction. The **engineering record** assigns mathematical types, units, assumptions, interfaces and tests. When these differ, the difference is made explicit in the claim register. The second record does not retroactively alter the first.

Five labels are used throughout:

Label	Meaning in this edition
Source	Directly stated or visibly represented in S1 or S2.
Definition	A mathematical convention introduced to make a statement precise.
Derived	A consequence proved from declared definitions or the source’s productions.
Proposal	An implementation choice or application extension added in this edition.
Demonstrated	A property checked by the accompanying code and recorded tests.

A source claim can be faithfully recorded without becoming a demonstrated claim. Conversely, a proposed interpreter can be useful without being represented as recovered author code.

1.3 Source hierarchy and preservation

Original files are retained byte-for-byte in `sources/originals/`. Their SHA-256 digests and sizes are recorded in `sources/source_index.json`. Derived text extractions are convenience copies; the original PDF remains authoritative for its equations, diagrams and page layout. Markdown line references use the actual original text lines, not an edited paraphrase.

The diagram on **S1 page 20** still places the golden-ratio tree around the cone/Klein construction. The diagram on **page 22** instead moves from word generation through grammatical expansion and topological recirculation. This is a documented architectural transition, not an instruction to require both mechanisms in the final implementation. The log-polar sketch on **page 7** also uses overlapping labels for pitch, phase, amplitude and harmonics; the later prose narrows phase to the angular coordinate and pitch to the radial scale. The ambiguity is recorded rather than hidden. [S1, pp. 7, 20, 22]

External primary sources, labelled R1–R9, are used to clarify established concepts and identify possible comparison methods. They do **not** validate this corpus as an integrated engine. No exhaustive prior-art investigation, legal registration or independent identity verification has been performed.

2. The corpus as stated

The following five layers preserve S2’s organization. Their assertions are reported as source content. The formal treatment and limitations appear in Sections 3–5 and Appendix A.

I. The data layer: 1-bit jitter and log-polar LUT

The source proposes low-bit-depth data with fine temporal organization. It describes a one-bit delta-sigma stream, calling its accompanying variation “dither” or “jitter,” and positions this as an alternative to heavy amplitude representations. A log-encoded polar lookup texture then organizes phase or timbre angularly and pitch or frequency radially. [S1, pp. 19–20; S2, lines 50–57]

The earlier musical example maps a key or note identifier to frequency, then advances a phase pointer through a waveform. That example supplies an intelligible precursor to the later multiscale representation: an input code does not itself contain a complete sound; it participates in a time-dependent interpretation. The corpus subsequently seeks to reduce or replace static tables through procedural construction. [S1, pp. 3–6]

Preserved intent: compact instructions and reusable structure should replace unnecessary expanded samples or coordinate arrays. **Unresolved in the source:** the stream decoder, table dimensions, channel meanings, numerical state and output clock.

II. Projective geometry and the Minkowski spacetime manifold

The phase loop is associated with an apex and a circle. Sweeping over time is associated with a cone; a faceted representation is associated with a pyramid; a spatially isotropic wavefront is associated with a sphere. Space and time are written as a direct sum,

$$M^4 = \mathbb{R}^3 \oplus \mathbb{R}.$$

The source then connects “SDF 0” with a null interval and the moving wavefront, using the quadratic expression

$$\Delta s^2 = \Delta x^2 + \Delta y^2 + \Delta z^2 - c^2 \Delta t^2.$$

The proposed engine thereby treats time as part of the representation rather than merely a playback parameter. [S1, pp. 17–20; S2, lines 59–72]

Preserved intent: geometric and temporal states should be organized coherently. **Unresolved:** which objects are slices, projections, approximations or coordinate domains; what metric is used; and whether the sound-cone language is a physical model or an analogy.

III. Non-orientable manifold: Klein bottle and blending

The source introduces a Klein-bottle construction to describe wrapping, inversion and return. It presents positive and negative states as cycling into one another and connects that circulation with the phrase “self-contained, stable topological engine.” It uses a smooth-minimum blend to describe merging fields:

$$\text{Blend}(a, b) = \min(a, b) - \frac{\max(k - |a - b|, 0)^2}{4k}.$$

The surrounding text calls this an exponential soft blend, although the displayed formula is polynomial. That discrepancy is preserved here and explicitly resolved only in the engineering record. [S1, pp. 14, 20–22; S2, lines 74–81]

Preserved intent: topological return and field blending are organizing primitives. **Unresolved:** a valid global distance representation, the relation between orientation and signal phase, and a dynamical law that could establish stability or energy retention.

IV. Replacing the tree: the 1-bit L-system grammar

The previous golden-ratio binary-search-tree stage is replaced by grammar-driven growth. The source's exact alphabet and production rules are

$$\Sigma = \{0, 1\}, \quad 0 \rightarrow 1, \quad 1 \rightarrow 01.$$

In its proposed geometric interpretation, 0 steps forward or propagates while incrementing time; 1 branches or rotates according to a golden-ratio relationship. The source uses $\phi \approx 0.618$, the reciprocal golden ratio. It claims that distance is implicit in word depth and length, and describes the string as the coordinate system. [S1, pp. 21–22; S2, lines 83–93]

Preserved intent: geometry is generated from a recipe rather than stored as an exhaustive grid.

Unresolved: the initial word, the exact angle, branch/return semantics, a metric decoder and the relationship between an instruction address and a physical coordinate.

V. Zero rasterization and structural output

The final layer states that the grammar can avoid voxelization and rasterization internally and be streamed as one-bit output. It names switching amplifiers, delta-sigma converters and acoustic or seismic devices as intended destinations. It also asserts perfect smoothness, freedom from aliasing and very small memory use. [S1, p. 22; S2, lines 95–99]

Preserved intent: a compact generative description should reach a physical or digital output without needless intermediate expansion. **Unresolved:** the instruction-to-signal encoding, reconstruction filter, device transfer function, finite precision, timing and measurement evidence.

The source-level synthesis

A compact binary recipe is intended to organize phase and scale, sweep geometric boundaries through time, wrap or invert states through a non-orientable construction, and produce output without a permanently expanded grid. The source proposes that grammar, geometry, topology and storage act as one coordinated description. The engineering question is which explicit interpreters and physical couplings make each part of that description valid.

2.1 Historical branches and expressive vocabulary

The earlier golden-ratio tree remains part of the corpus's history. Its proposed split values are associated with ϕ and ϕ^2 ; because $\phi + \phi^2 = 1$, these can define complementary proportions. That arithmetic does not by itself specify a balanced data structure, a golden-section optimization problem or a guarantee against unwanted resonances. The final grammar need not retain this tree. [S1, p. 20; S2, lines 83–93]

“Absolute zero,” “gestation,” “crush,” “event horizon” and “entropy” occur as expressive connections around a boundary or transition. They are retained as corpus vocabulary. None is silently converted into a thermodynamic law, a relativistic horizon or an amplitude measurement. The source's use of acoustic weaponization and intentional structural disruption is likewise retained as an **archival application branch**, not developed into operational designs here. [S1, pp. 8–15; S2, lines 137–149]

3. Semantic boundaries and mathematical types

3.1 A single symbol is not a single type

The corpus moves between descriptions that are related but not interchangeable. A musical signal may be represented as a continuous function, as a vector of samples, as oscillator state or as a symbolic score. Calling a note a wave rather than a vector in the opening discussion distinguishes a physical phenomenon from one representation; it does not prohibit vector-space representations of signals. [S1, pp. 2–4]

Similarly, a one-bit **instruction** can mean “move” or “rotate,” while a one-bit **modulation sample** selects a signal level. A binary **chart-parity flag** identifies orientation relative to a chosen lift. These bits have different meanings even when they share the same storage type.

Object	Formal role	Units or domain
b_j	Grammar instruction	$\{0, 1\}$
w_h	Finite word after h rewrites	Σ^*
θ or α	Phase or geometric heading, separately named	Radians
ρ	Log-frequency coordinate	Octaves relative to a reference
x, t	Physical position and time	Length, time
$d(x, t)$	Spatial distance to a specified boundary	Length
$q(x, t)$	Acoustic quadratic interval	Length squared
g	General scalar/control field	Declared per field
$p(x, t)$	Physical pressure, when modeled	Pressure
(u, v, σ)	Quotient-chart coordinates and lift orientation	Dimensionless; $\sigma = \pm 1$
a_n and β_n	Sample amplitude and quantizer output bit	Normalized amplitude; $\{0, 1\}$

A typed implementation must reject or explicitly convert incompatible objects. A value in octaves is not a physical distance. A quadratic interval is not an SDF merely because both can equal zero.

3.2 Five different meanings of combination

Signal addition combines compatible amplitudes. In a declared linear pressure model, $p = p_A + p_B$ represents superposition. **Geometric union** can be encoded by the nonpositive sublevel set of $\min(d_A, d_B)$ under a negative-inside convention. **Smooth minimum** changes the combined implicit field near a junction. **Direct sum** constructs a product-like vector-space decomposition with unique component coordinates. **XOR** is a Boolean operation on bits. None may replace another merely because the source uses “blend” or the symbol $\oplus$ nearby. [S1, pp. 13–17]

For example, equal binary inputs XOR to zero, but that fact does not establish cancellation of arbitrary acoustic waves. Likewise, an internal direct-sum decomposition has component subspaces intersecting trivially; it does not imply that independent physical fields meet only at a geometric zero boundary. These are type distinctions, not a rejection of the corpus’s interest in a common architecture.

3.3 A scalar field is not automatically a distance field

This edition reserves **signed distance** for a field whose magnitude equals distance to a specified boundary under a specified metric, with a declared side convention. A generic implicit field can share its zero set without sharing its values. A distance bound is another distinct object. Hart's treatment explicitly separates these concepts and makes the required bounds visible in geometric evaluation. [R6, §2.1]

Changing to polar or logarithmic coordinates does not automatically preserve Euclidean distance. A "log-polar SDF" must state whether its values measure distance in physical space, in a chart metric or in an abstract parameter space. Where that information is absent, the correct general type is **implicit field**, **control field** or **level-set function**.

3.4 Signal continuity, finite sampling and precision

A continuously defined function can still generate sampled output with finite timing and arithmetic error. A texture interpolator uses finitely stored values and a declared reconstruction rule. An instruction grammar does not by itself specify a passband or a reconstruction filter. Delta-sigma conversion has separate oversampling, noise-shaping and filtering stages, as the architecture comparison in R2, Figure 3, makes explicit. [R2, PDF p. 5]

Amplitude dither and clock jitter must also remain separate. Dither is an added perturbation chosen for a particular quantization purpose; jitter is uncertainty or variation in sample timing. Their effects are not interchangeable. The accompanying code implements neither dither nor timing-jitter simulation. [R3; R4]

4. An explicit engineering formalization

Status of this section: definitions and proposals, except where a production or formula is explicitly identified as source-derived. The purpose is to make the corpus implementable without claiming that missing choices were already specified.

4.1 Engine signature and data flow

Define an engine instance by

$$\mathcal{E} = (G, Z, T, A, F, B, O, \Lambda), \quad G = (\Sigma, w_0, P). \quad (\text{E1})$$

Here G is the grammar, Z the typed state space, T the instruction-transition family, A the address/chart map, F the collection of evaluable fields, B the permitted typed combination operators, O the output adapters, and Λ the parameters, units and resource limits. The distinction between A and F prevents an address from being mistaken for a measured physical field.

For a finite generation and its symbols,

$$w_h = P^h(w_0) = b_0 b_1 \cdots b_{N_h-1}, \quad z_{j+1} = T_{b_j}(z_j; \Lambda), \quad (\text{E2})$$

$$\xi_j = A(z_j), \quad g_j = F(\xi_j), \quad a_n = O(\{z_j, g_j\}, t_n; \Lambda). \quad (\text{E3})$$

The conceptual order of S2 describes how the author introduces the ideas. Equations E1–E3 instead describe an **execution order**: generate instructions, update state, query a representation, then produce an output. This is an explicit engineering organization, not a claim that S2 already supplied these interfaces.

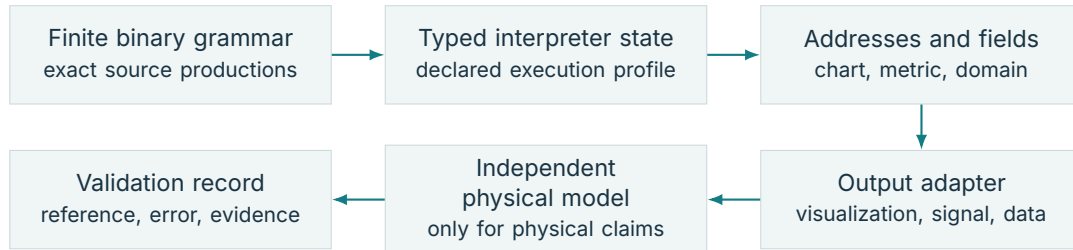


Figure 1. Proposed execution architecture. A physical model is conditional: a symbolic or visual output need not traverse that block. Topological indexing belongs to the address/field layer; it is not a substitute for a physical model.

A conforming experiment records its grammar, finite depth, initial state, interpretation profile, coordinate conventions, field definitions, output clock and resource budget. The same word with a different interpreter is a different experiment.

4.2 Grammar, finite generations and streaming

Use the source productions exactly:

$$P(0) = 1, \quad P(1) = 01, \quad P(ab) = P(a)P(b). \quad (\text{E4})$$

The **reference profile chooses** $w_0 = 0$; the source does not explicitly fix that seed. Parallel re-writing gives

$$0, \quad 1, \quad 01, \quad 101, \quad 01101, \quad 10101101, \dots$$

These particular finite words are not assumed to be successive nested prefixes of one infinite word. A streaming request therefore names a finite generation h and, optionally, a prefix length within that generation. This avoids silently changing the source's substitution to a more familiar ordering.

Grammar generation and geometric interpretation remain separate. That separation is also explicit in the primary L-system account: rules determine rewritten words, while a turtle interpretation adds positions, headings and step parameters. [R1, §§1.2–1.3]

The demonstrator uses an explicit stack rather than materializing intermediate generations. A depth limit and output budget prevent accidental unbounded expansion. This reduces auxiliary storage; it does not eliminate the work needed to emit the requested symbols.

4.3 Interpreter profile: move or rotate

For the demonstrator, let the interpreter state on a two-dimensional universal cover be

$$z_j = (u_j, v_j, \alpha_j, \tau_j), \quad (\text{E5})$$

where u, v are dimensionless cover coordinates, α is a heading and τ is **grammar time**. Choose positive step ℓ , positive grammar-time increment $\Delta\tau$, and a declared turn angle γ . Then

$$T_0 : \begin{cases} u_{j+1} = u_j + \ell \cos \alpha_j, \\ v_{j+1} = v_j + \ell \sin \alpha_j, \\ \alpha_{j+1} = \alpha_j, \\ \tau_{j+1} = \tau_j + \Delta\tau, \end{cases} \quad T_1 : \begin{cases} u_{j+1} = u_j, \\ v_{j+1} = v_j, \\ \alpha_{j+1} = (\alpha_j + \gamma) \bmod 2\pi, \\ \tau_{j+1} = \tau_j. \end{cases} \quad (\text{E6})$$

In E6 the first coordinate is u ; it is an ordinary coordinate, not a propagation-speed symbol. The source’s “branch or rotate” is underdetermined. This profile chooses **rotation only** and

$$\phi = \frac{\sqrt{5} - 1}{2}, \quad \gamma = 2\pi(1 - \phi) = 2\pi\phi^2. \quad (\text{E7})$$

E7 is an introduced angle convention, not a recovered author parameter. A branching profile would need a state-fork operation, branch identifiers, return/merge rules and branch budgets. Calling 1 a branch without specifying those mechanisms would not implement a branching engine.

Grammar time does not wrap with the topological coordinates. The audio example has its own sample clock $t_n = n/f_s$ and a declared rescheduling of grammar events across that clock. Topological recurrence must not silently reset physical time or imply access to future states.

4.4 Phase and logarithmic coordinates

For positive frequency and reference frequency, define

$$\rho = \log_2(f/f_{\text{ref}}), \quad f = f_{\text{ref}}2^\rho. \quad (\text{E8})$$

An octave increase is **addition of one** to ρ . If a separate linear radius R is proportional to frequency, doubling R can represent an octave; but R and ρ are not the same coordinate. This resolves the source’s two radius descriptions by naming two conventions instead of treating them as identical. [S1, p. 7; S2, line 57]

For a spatial log-polar chart, separately define

$$r = \sqrt{x^2 + y^2} > 0, \quad \eta = \ln(r/r_0), \quad \theta = \text{atan2}(y, x). \quad (\text{E9})$$

The chart is undefined at the origin. The source’s apex therefore requires its own state or chart handling; it cannot simply be fed into a logarithm. The reference code rejects that undefined query.

A normalized oscillator phase can advance according to

$$\psi_{n+1} = (\psi_n + f_n/f_s) \bmod 1. \quad (\text{E10})$$

A later float-free implementation could use a B -bit phase accumulator and a rounded integer increment. Under nearest rounding, its frequency-step size would be $f_s/2^B$, not infinite precision. Such a port is a future numerical profile. The supplied implementation uses floating point and labels that departure from the all-one-bit ambition explicitly.

LUT option. A table backend must declare dimensions, channel meanings, address decoding, wrap rules and interpolation. An analytic backend can evaluate the chosen field without storing a dense table. Neither backend eliminates state: a single incoming bit needs history or decoding to select among more than two addresses.

4.5 Space, time and the sound cone

Let physical positions lie in $X = \mathbb{R}^3$, time in $T = \mathbb{R}$, and let an event be (x_0, t_0) . A direct-sum notation organizes coordinates; a metric or quadratic form is additional structure. Minkowski space requires that extra structure, as distinguished in the relativity account in R5. [R5, PDF p. 20]

For an explicitly chosen **homogeneous, stationary, isotropic medium** and positive propagation speed c_s , use the acoustic analogue

$$q(x, t) = \|x - x_0\|^2 - c_s^2(t - t_0)^2. \quad (\text{E11})$$

For $t \geq t_0$, the spherical front is

$$\Gamma_t = \{x : \|x - x_0\| = c_s(t - t_0)\}, \quad d_t(x) = \|x - x_0\| - c_s(t - t_0). \quad (\text{E12})$$

At a positive fixed elapsed time, d_t is the signed spatial distance to that sphere. At the initial instant the front degenerates to a point. E11 and E12 have a common future zero set but different dimensions and off-boundary values. This is a controlled acoustic model, not a claim that a material sound field is relativistic Minkowski spacetime.

A circle swept in one time direction describes a two-spatial-plus-one-temporal construction. A sphere swept in time gives the three-spatial-plus-one-temporal front family above. A pyramid can be a faceted approximation only after its construction and error criterion are specified. None of these dimensional changes follows from naming the objects alone.

Boundary is not amplitude. A linear homogeneous pressure model might independently be written as

$$\partial_t^2 p - c_s^2 \nabla^2 p = s(x, t), \quad (\text{E13})$$

together with initial conditions, boundary conditions and an actual source. E12 provides a geometric front in its assumptions; it does not solve E13 for pressure, reflections, absorption or caustic amplitude. Independent acoustic models explicitly include equations and medium parameters. [R8]

4.6 A valid topological indexing proposal

For the non-orientable component, define an abstract Klein-bottle indexing surface by the quotient

$$K = \mathbb{R}^2 / \sim, \quad (u + 1, v) \sim (u, -v), \quad (u, v + 1) \sim (u, v). \quad (\text{E14})$$

The quotient is a **two-dimensional indexing space**. It is not an invertible conversion of arbitrary four-dimensional spacetime into a two-dimensional surface. An address map from a larger state space to K can be many-to-one, and must say what information it discards.

A canonical representative for a chosen lift is

$$m = \lfloor u \rfloor, \quad U = u - m, \quad V = ((-1)^m v) \bmod 1, \quad \sigma = (-1)^m. \quad (\text{E15})$$

The flag σ records orientation relative to that lift and chart. Equivalent quotient positions may have different lift-orientation labels. For a heading stored on the cover, the corresponding local heading reflects as $\alpha_{\text{local}} = \sigma \alpha$ modulo 2π .

A well-defined ordinary scalar field on this quotient must respect the identifications. The demonstrator uses the deliberately simple example

$$g_K(u, v) = \frac{1}{2} \{ \cos(2\pi u) + \cos(2\pi v) \}. \quad (\text{E16})$$

It satisfies $g_K(u+1, -v) = g_K(u, v)$ and $g_K(u, v+1) = g_K(u, v)$. It **does not** flip pressure sign at a seam. An orientation-sensitive or twisted state could be defined with a different transformation law, but that would be an additional model.

The conventional Klein bottle is non-orientable and cannot be embedded in ordinary three-space without the familiar immersed representation's self-intersection. A global inside/outside SDF for that imagined embedded surface therefore cannot simply be assumed. The quotient-chart proposal preserves the source's interest in twisted return while explicitly replacing its unspecified global SDF construction. [R7, §1.2]

4.7 Blending contract

For compatible scalar fields with common units and fixed $k > 0$, retain the source formula exactly:

$$B_k(a, b) = \min(a, b) - \frac{\max(k - |a - b|, 0)^2}{4k}. \quad (\text{E17})$$

The implementation may evaluate an algebraically equivalent normalized expression to avoid unnecessarily squaring large values. This is a **polynomial smooth minimum**, not the exponential function described in the source's prose. It is a field-construction operator; it is not automatically a wave-superposition rule, a metric-preserving union or a physical recirculation operator. [S1, p. 14; S2, lines 79–81]

The resulting field should be typed as an implicit blend unless its distance properties are separately established. If a downstream routine requires conservative distance bounds, the blend and every coordinate transform must supply the needed bound or be excluded from that routine.

4.8 Output adapters and the one-bit distinction

For sound design, a bounded normalized waveform can be constructed from phase and control values. That is an **output adapter**, not a claim that a topological scalar is physical pressure. The example file uses grammar-controlled frequency and a two-harmonic morph with a start/end envelope. It is saved as a conventional reference WAV without automatic playback.

A separate demonstrator implements first-order error feedback for input samples $a_n \in [-1, 1]$:

$$v_n = e_{n-1} + a_n, \quad \beta_n = \begin{cases} 1, & v_n \geq 0, \\ 0, & v_n < 0, \end{cases} \quad q_n = 2\beta_n - 1, \quad e_n = v_n - q_n, \quad e_{-1} = 0. \quad (\text{E18})$$

The grammar symbols b_j and modulation bits β_n are different sequences. E18 is an offline algebraic example, not a complete converter or a power-electronics driver. It does not automatically oversample its input, add dither, simulate clock jitter or reconstruct analog output. R2 supplies the broader converter context; it does not establish the quality of this example. [R2]

The exported bitstreams are files for inspection. No gate timing, amplifier control, directional-acoustic optimization or transducer-driving interface is included.

5. Derived properties and decisive checks

These results are derived from the declared productions or definitions. Their scope is deliberately narrower than the source’s strongest performance language.

D1. Fibonacci growth and the reciprocal golden ratio

Let z_h, o_h be the zero and one counts after h rewrites. Each old zero contributes one new one; each old one contributes a zero and a one. Therefore

$$\begin{pmatrix} z_{h+1} \\ o_{h+1} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} z_h \\ o_h \end{pmatrix}. \quad (\text{D1})$$

With $w_0 = 0$, the counts start at $(1, 0)$. For $h \geq 1$, $z_h = F_{h-1}$ and $o_h = F_h$, where $F_0 = 0, F_1 = 1$. The word length is $N_h = F_{h+1}$, including $N_0 = 1$. The count ratio converges to

$$\frac{z_h}{o_h} \rightarrow \phi = \frac{\sqrt{5} - 1}{2}.$$

This establishes the source’s count relationship under the declared seed. It does not establish a resonance theorem, a search-complexity guarantee for arbitrary data, or an optimal geometry.

D2–D3. Address capacity and logarithmic shift

A single isolated bit has two possible values. Addressing a table with more than two entries therefore needs multiple bits, an external selector, accumulated state or another decoding rule. The phrase “direct indexing” is incomplete until that rule is specified.

From E8,

$$\log_2(2f/f_{\text{ref}}) = 1 + \log_2(f/f_{\text{ref}}).$$

Thus the octave rule is exact for the defined coordinate, and round-trip tests can check its numerical implementation. It does not make interpolation in that coordinate exact for an arbitrary waveform.

D4. Shared zero is not equality of fields

Write $r = \|x - x_0\|$ and $\Delta t = t - t_0 \geq 0$. Then

$$q = (r - c_s \Delta t)(r + c_s \Delta t) = d_t(r + c_s \Delta t). \quad (\text{D2})$$

The future zero set agrees, but at normalized $r = 4, c_s \Delta t = 3$ the distance is 1 while the quadratic expression is 7. The code checks both the shared zero and the unequal off-boundary values. Consequently, the source’s “SDF 0” can name a shared boundary in this example without making its SDF and interval numerically identical.

D5–D6. Smooth minimum moves an original boundary

If $|a - b| \geq k$, E17 reduces to $\min(a, b)$. If $a = b = 0$, however,

$$B_k(0, 0) = -k/4. \quad (\text{D3})$$

The point formerly on both zero sets is no longer a zero of the blended field. More generally $B_k(a, a) = a - k/4$, so the operation is not idempotent. These direct substitutions refute a blanket assertion that the displayed blend always preserves the original absolute-zero boundary. They do not make the blend useless: changing the junction is its intended geometric effect.

For negative-inside fields, the nonpositive set of $\min(d_A, d_B)$ represents the union of the two non-positive sets. Exact distance values, and even the interpretation of degenerate internal zero seams, need separate care. Neither $\min$ nor smin substitutes for addition of physical pressure samples.

D7. Topological orientation is not scalar phase

Equation E16 is a counterexample to automatic sign inversion: it is a nontrivial ordinary scalar on the declared Klein quotient and returns the same value at identified points. The chart orientation changes under an odd horizontal wrap, while the scalar does not acquire a mandatory minus sign. This is why the code stores orientation separately from amplitude.

D8. A word length cannot uniquely determine distance

Under E6, 00 and 01 have equal length but different final positions and grammar times. A scale change in ℓ also changes their geometry without changing the word. Therefore word length and derivation depth alone cannot define distance to a geometric boundary. A valid address-to-distance system needs at least the interpreter, initial state, geometry and metric. The useful research proposition is **distance recoverable from a compact recipe and decoder**, not distance supplied by word length in isolation.

D9. Compression and execution have different costs

A fully packed N -symbol binary word occupies $\lceil N/8 \rceil$ bytes, ignoring metadata. Its rule-and-seed description can be much shorter. The reference depth-first generator uses auxiliary stack space proportional to derivation depth for a fixed seed; emitting a full finite generation still takes time proportional to the number of symbols, up to the initial depth traversal. Saving every interpreted state again takes output-proportional storage.

A useful accounting identity is

$$M_{\text{total}} = M_{\text{rules}} + M_{\text{interpreter}} + M_{\text{state}} + M_{\text{fields}} + M_{\text{buffers}} + M_{\text{saved output}}. \quad (\text{D4})$$

“Thirty packed bytes” for a word is not “thirty bytes for the engine.” The supplied Python program deliberately favors inspectability over minimum embedded memory.

D10. Raw grammar is not a general audio modulation code

For the source grammar, $o_h/N_h \rightarrow \phi$. Mapping its bits directly to bipolar values $2b - 1$ therefore gives an asymptotic mean

$$2\phi - 1 \approx 0.236068, \quad (\text{D5})$$

not zero. It cannot represent an arbitrary zero-mean target waveform without another encoding. This calculation is a concrete reason to separate grammar instructions from signal quantization.

For E18, by contrast, $|e_n| \leq 1$ follows by induction for $a_n \in [-1, 1]$. Since

$$q_n - a_n = e_{n-1} - e_n,$$

summing N samples telescopes and gives

$$\left| \frac{1}{N} \sum_{n=0}^{N-1} (q_n - a_n) \right| \leq \frac{1}{N}. \quad (\text{D6})$$

This is a mean-tracking bound for this recurrence and initialization. It is **not** a bound on audible noise, instantaneous reconstruction error or physical pressure. The code tests the state bound, a DC case and a varying input.

6. Reference implementation and recorded results

6.1 What is supplied

The engine/ package is a standard-library Python research demonstrator. It is newly prepared implementation code, not source code extracted from the author's files. The exact source productions and polynomial blend are retained. The seed, step length, turn angle, quotient convention, example scalar, sample scheduling and output mapping are explicitly introduced choices.

The implementation contains a bounded grammar generator, exact count evaluator, pitch encoder/decoder, spatial log-polar transform, phase update, sphere distance and acoustic-interval checks, polynomial smooth minimum, a Klein quotient chart, a move/rotate interpreter, a first-order one-bit quantizer and binary packing utilities.

The demo writes a grammar word and packed grammar bits, a trace of its interpreted states, growth and pitch tables, field-check values, a conventional reference WAV, a separate quantizer bitstream, a signal preview and a JSON results record. The WAV is **not** reconstructed from the one-bit file; the two are separate outputs.

6.2 Reproduction commands

Run these commands from the extracted package root with Python 3.10 or later. The recorded run used Python 3.13.5.

```
python tools/verify_package.py
python -m unittest discover -s tests -v
python -m engine.demo --output ../klootwijk_demo_run
```

The first command checks the delivered files against the manifest. The second runs the tests. The third regenerates examples in a separate sibling directory, preserving the original result record. Regenerated floating-point output and environment metadata may differ across runtimes.

6.3 Recorded results

Check or output	Recorded value	Interpretation
Automated tests	35 passed	Listed software checks only
Grammar depth / seed	12 / 0	Chosen reference profile
Emitted symbols	233	Exact finite word
Zero / one counts	89 / 144	Ratio 0.6180556
Packed grammar bytes	30	Excludes program, state and metadata
Negative-orientation trace events	118	Chart parity, not negative pressure
Reference WAV	16,000 samples at 16,000 Hz	One-second offline multibit example
Reference digital peak	approximately 0.09370	Normalized value, not an acoustic safety limit
Quantizer maximum absolute state	1.0	Consistent with E18's bound
Mean tracking error	approximately 9.17×10^{-7}	Below $1/N = 6.25 \times 10^{-5}$

Full machine-readable values are in `examples/demo_summary.json`; the test log is in `examples/test_results.txt`. The runtime in that log is incidental to this environment and is not a comparative performance benchmark.

6.4 What is not implemented or demonstrated

There is no branch-stack interpreter, texture backend, float-free arithmetic port, globally sided Klein-bottle distance oracle, caustic-amplitude solver, general physical propagation solver, dither model, jitter model, analog reconstruction filter or hardware driver. There is no evidence here of physical focusing, stored-energy recirculation, universal resonance avoidance, zero aliasing or production readiness.

The example quantizer is run at the stated sample clock for inspection; no claim is made that its oversampling ratio or reconstruction quality is suitable for a finished audio product. The saved normalized WAV amplitude cannot guarantee a listening level because playback hardware and gain are outside the model. The package neither plays the file nor connects to devices.

6.5 Portability and resource boundaries

Grammar words, counts and packed bit order are exact for the declared finite inputs. Floating-point trigonometric values may differ slightly across runtimes and platforms; tests use numerical tolerances where appropriate. Cross-platform bit-identical audio would require an additional arithmetic and rounding specification.

The public functions reject malformed words, non-finite numeric inputs, invalid radii, an undefined log-polar origin and excessive output requests. The design is intentionally bounded. Removing a budget should be an explicit experiment change, not an accidental route to “infinite” generation.

7. Broad-scale application portfolio

All application assessments are proposals or inferences from reusable components, not measured market claims or established performance of the complete engine. “Software prototype” means a sensible next experiment; it is not an external maturity certification. Broad applicability does not require every sector to use every layer. A music system may use grammar and log pitch without a Klein chart; a geometry system may use fields and grammar without any audio output.

7.1 Creative and interactive systems

A01 — Procedural music and sound design. The direct fit is a grammar-controlled synthesizer or sound-design environment. Binary instructions can schedule changes in pitch, phase, timbre and envelopes, while a field supplies a structured modulation source. This continues the source’s original note-to-frequency and procedural-wave discussion. The baseline should be an ordinary deterministic sequencer using the same oscillator or wavetable, so any advantage comes from the representation rather than an unequal signal backend. Measure unwanted spectral components, control latency, total memory and creative controllability. The included WAV demonstrates one mapping, not a complete instrument. [S1, pp. 3–8, 21–22; R1]

A02 — Audiovisual instruments, installations and immersive media. A common state can drive animation and sound without maintaining unrelated timelines. The source already mentions shader and visualizer directions; immersive or XR content is an added application inference. A practical prototype would export one event trace to two renderers and measure their synchronization, repeatability and frame/audio cost. It need not claim that the visual field is an acoustic simulation. Compare against a shared conventional timeline with equivalent events. [S1, p. 8]

A03 — Mathematics and physics education. The corpus offers a connected teaching sequence: re-write rules, count ratios, interpretation, log coordinates, boundary sets, cone slices, quotient charts and signal encoding. The strongest educational use is to make distinctions visible, including why a shared zero set is not field equality and why orientation is not pressure sign. The package already contains inspectable examples. A future teaching interface should be evaluated for correctness and understanding, not presented as experimental proof of all physical claims. [Concepts: S1, pp. 13–22; educational use proposed; R1, R5, R7]

A10 — Haptic and accessible interaction design. A related but additional proposal is to map bounded states to tactile or multimodal cues. The potential value is repeatable, parameterized patterns and shared timing across modalities. A first version should remain an offline interaction model. Any device study needs separate actuator limits, comfort assessment and user evaluation. This is not a medical, therapeutic or neurological claim. Compare against simpler cue sequences before assuming that topology adds value. [Proposal based on signal/control components]

7.2 Geometry, representation and computation

A04 — Embedded signal or control sequencers. The source explicitly targets small computing platforms. The implementable proposition is not “zero memory,” but replacement of long stored event lists with a compact recipe and a bounded interpreter. A later fixed-point port could quantify program size, state size, buffers, worst-case latency and numerical error. Compare against a stored event table and a conventional phase accumulator at equal output quality. Streaming moves cost from storage into computation; the measurement should reveal whether that trade is useful on the chosen target. [S1, pp. 3–6, 21–22; S2, lines 151–156]

A05 — Implicit geometry and CAD exploration. Grammar-controlled field primitives could generate families of conceptual shapes and visualize how blended boundaries change. The useful output is an inspectable model with known units and error, not an assumption that every blend remains an exact SDF. Compare with constructive solid geometry and mesh-based representations. Validate topology, boundary deviation, distance-query error and computation time. Production manufacturing or tool control would be a separate engineering programme; the present proposal is shape exploration and non-production path visualization. [S1, pp. 13–14, 21–22; R6]

A06 — Compact procedural assets and replay. Store or transmit the grammar, seed, decoder version, parameters and any residual data, then regenerate the agreed output. The fit includes audiovisual presets, animation recipes, procedural scenes and reproducible state traces. Compare total encoded size and decode cost against ordinary compression and existing procedural formats. The decoder cannot be treated as free storage. Arbitrary source material is not guaranteed to have a short grammar, and different arithmetic implementations may need a shared conformance profile. [S1, pp. 21–22; application extension proposed]

A09 — Multiscale vision and descriptor research. S1 explicitly connects log-polar textures with scale/rotation-oriented visual computing. The proposed extension is an experiment in image registration, descriptor organization or procedural addressing. Begin with a conventional log-polar baseline using identical input images and interpolation. Measure registration error, robustness to transformations, chart distortion and compute cost. No image-recognition accuracy or scale invariance for this engine has been demonstrated, and this edition does not supply a computer-vision benchmark. [S1, pp. 7–8]

A12 — Reproducible scientific and creative workflows. A versioned recipe can bind an experiment’s instructions, interpretation, units and outputs together. That is useful for replay and audit even without a new physical effect. The companion package is an initial example: it records source provenance, definitions, code, tests and hashes. A future format could preserve many projects with explicit decoder versions and compatibility rules. Compare it against straightforward versioned scripts and data rather than assuming a grammar is always the simpler representation. [Workflow application proposed]

7.3 Physical-model and research applications

A07 — Wave-simulation front ends and room-acoustics studies. The engine can be investigated as a source-schedule generator, geometry description or parameter interface for an independent wave solver. This preserves the corpus’s spatial and temporal organization without mistaking an SDF for a pressure solution. For example, a study could send identical source and geometry parameters through the engine’s adapter and a conventional setup interface, then use the same solver. Differences should concern setup cost, representation size or control, not an unsupported change in physical laws. [S1, pp. 8, 10–11, 18–22; R8]

A08 — Non-destructive sensing and tomography. Acoustic, seismic and inspection settings are named in the source. The benign research fit is candidate signal/model encoding and parameterization of an inversion workflow. A physical measurement programme still needs calibrated sources and receivers, medium information, noise characterization and an identifiable inverse problem. Evaluate reconstruction error and uncertainty against established methods using matched observations and compute budgets. The grammar does not itself reveal subsurface structure or replace measured data. Intentional structural disruption is excluded from development. [S1, pp. 9–10; S2, lines 144–149]

A11 — Simulation-based inverse design. A grammar can define a constrained family of geometries or control schedules, with an external optimizer selecting parameters against a separately implemented simulation objective. This is an extension, not a mechanism already specified in the corpus. Modular differentiable wave simulation provides an example of the kind of external infrastructure such a study could use. Compare against standard parameter families with the same solver, optimizer and evaluation budget. Report constraint violations and held-out validity, not only an optimized training objective. [R9; coupling to this corpus proposed]

7.4 Prioritization and adoption logic

The recommended **first demonstration track** is an inspectable procedural sound-and-geometry notebook built on A01, A03 and A06. These uses can be tested without assuming an unprovided physical model. The **second track** is a measured embedded or audiovisual port, A04 or A02, after a conventional baseline is available. The **research track** is A07, A08 or A11, with an independent solver and domain-specific validation.

Priorities reflect dependency order, not commercial demand. The twelve-entry CSV/JSON matrix records source basis, baselines and metrics.

8. Physical fit, scale and architectural limits

8.1 Three different senses of “fits physically”

As information represented on physical hardware: a grammar interpreter, state machine and signal adapter can be implemented using conventional digital hardware. That is a realizable computational direction, although this package only executes a software reference.

As a description of a physical model: fields, source schedules and parameters can be supplied to a wave model with declared assumptions. The correspondence then has to be checked against the governing equations and measurements. A compact encoding may simplify representation without reducing the solver’s physical obligations.

As a claim about a new physical mechanism: the corpus does not yet supply a governing derivation or experiment establishing automatic energy retention, arbitrary beam invariance, exact caustic amplitude from distance alone or universal avoidance of harmful resonances. Those are separate hypotheses. They cannot be obtained by re-labelling an instruction stream as a wave.

The constructive path is to retain the first two meanings and formulate each stronger mechanism as a testable claim with explicit inputs, outputs and failure conditions.

8.2 What the geometry does and does not determine

A boundary can locate a surface or a wavefront in a specified model. It does not determine pressure, phase coherence, dissipation, material response or measurement uncertainty without more information. Near a caustic, a chosen geometric description may be useful for organizing the calculation; it is not itself an amplitude solver.

S1 names a “Normal Caustic Coordinate System” and mentions Airy/asymptotic language, but provides neither a reproducible coordinate definition nor coefficients and boundary conditions that would turn those phrases into a complete algorithm. The external check in this edition did not establish that exact named construction as the corpus’s specified solver. Its status is therefore **source terminology with an unresolved implementation**, not a validated subsystem. [S1, pp. 10–11]

A reference pressure model must be chosen independently. k-Wave documents coupled equations, material heterogeneity and absorption rather than inferring them from a geometry label; j-Wave likewise supplies an explicit simulation framework. They are comparison resources, not endorsements of the proposed engine. [R8; R9]

8.3 Recurrence, stability and energy

A point returning to an equivalent chart coordinate is a statement about representation. A physical system retaining or dissipating energy is a statement about dynamics. The two must be connected by an explicit physical realization before the former can imply anything about the latter.

For a proposed realization, record an energy balance such as

$$E(t_2) - E(t_1) = W_{\text{in}} - E_{\text{loss}} - E_{\text{radiated}}, \quad (\text{P1})$$

with each term defined for that actual system. P1 is an accounting requirement proposed here; it is not a derived energy law for a Klein-bottle engine. A feedback implementation would also need a defined update law, delays, gain, saturation and a stability analysis. Computational wrapping alone supplies none of those quantities.

The minimum meaningful experiment would compare the same realized system with and without the proposed topological/state feedback, under matched inputs, while measuring stored and outgoing energy. No such experiment is part of the supplied corpus or the accompanying software results.

8.4 Scaling from a small program to a broad platform

Storage scaling. A grammar can compress regular structure; output, state and decoder costs remain. Texture storage scales with the number of table entries, channels and bits per channel. Removing a table replaces that cost with procedural evaluation rather than making it vanish.

Execution scaling. Full expansion of the specified grammar has Fibonacci growth in the derivation depth. Streaming can avoid retaining the full expansion, but a request to emit all symbols still grows with its length. Random access to a symbol can use subtree lengths; random access to the interpreted state additionally needs state-transition summaries or replay. It is not automatically a logarithmic-time distance query.

Parallel scaling. Independent field queries or explicitly summarized grammar subtrees can be candidates for parallel execution. Stateful interpretation may remain order-dependent because moving and rotating need not commute. A GPU, distributed worker or cache is an optional backend, not a proof of speedup.

Representation scaling. “No grid” can describe how a model is stored. A renderer, signal output or numerical solver may still sample that model. Samples per output interval, error tolerance and backend complexity determine the real resource cost.

Organizational scaling. A shared kernel should stabilize the grammar, type system, profile format and provenance. Domain adapters should own their physics and validation. This is a more testable platform boundary than claiming that one scalar representation already solves every domain.

8.5 Boundaries on development

The source’s harmful acoustic and structural-damage applications remain documented only at a non-operational level in the analysis, with originals preserved for provenance. This edition develops no weaponized output, distress-maximizing signal, destructive resonance control or power-stage instructions.

Benign physical studies should begin with simulation and independently reviewed, calibrated measurement arrangements. The package contains no device interface and establishes no safe physical output level. Its general structure is a computational research framework, not a deployment authorization.

9. Validation programme and decision gates

A useful validation programme separates **conformance to the recipe**, **numerical correctness**, **comparative engineering performance** and **physical efficacy**. Passing an earlier gate does not automatically pass a later one.

Gate 0 — Freeze the semantic profile

Record the productions, seed, finite-generation convention, meanings of 0 and 1, clocks, step and angle, chart identifications, field types, metric units and output adapter. Make the tree-to-grammar transition explicit. Decide whether the implementation is exploratory floating point or a defined finite-word arithmetic profile.

Evidence required: a versioned specification and configuration. **Current status:** a declared reference profile is supplied; the source-level ambiguities remain visible rather than being erased.

Gate 1 — Exact symbolic and local numerical checks

Use exact fixtures for rewritten words, counts and packed bytes. Test invalid inputs and resource limits. Test pitch round trips, origin handling, phase wrap, smin behavior, cone zero sets, chart seams and scalar invariance. A test that checks a known counterexample is as valuable as a test that confirms an intended property.

Evidence required: repeatable automated tests. **Current status:** 35 tests pass in the recorded environment. The tests establish their finite cases and algebraic invariants; they do not establish correctness of an unimplemented full engine.

Gate 2 — Numerical convergence and representation fidelity

For any field advertised as an exact SDF, compare its value against known distances and inspect the local residual $||\nabla d|| - 1|$ where differentiable. Such a residual is a diagnostic, not by itself a global proof. For a chart transform, compare physical and chart distances under the declared metric. For sampled signals, measure spectrum, reconstruction error and behavior as precision and sample rate change.

Evidence required: saved inputs, analytic or high-accuracy reference results, tolerances and convergence plots. **Current status:** elementary local checks are supplied; no full distance-oracle, audio-quality or physical-solver convergence campaign is supplied.

Gate 3 — Matched-baseline engineering comparison

Measure the same task at the same output quality against simpler alternatives. For A01 use the same oscillator backend. For A04 include code, state and buffer memory. For A06 count decoder and metadata size. For A07 use the same physical solver. Record both favorable and unfavorable cases.

Evidence required: repeatable workload definitions; total memory; worst-case and distributional latency; output error; build settings; hardware details. **Current status:** not performed. The execution time in the unit-test log is not such a benchmark.

Gate 4 — Physical validation for a selected benign use

Specify the medium, devices, transfer functions, initial and boundary conditions, calibration method and observable quantities. Run repeated measurements and compare them to predictions and a baseline. For sensing, include reconstruction uncertainty and identifiability. For energy claims, account for input, storage, loss and radiation.

Evidence required: independently interpretable measurement data and a model that can be falsified. **Current status:** no physical measurements are supplied or generated in this edition.

Gate 5 — Application-specific integration

Only after the earlier gates should a selected adapter be assessed for device compatibility, reliability, user needs and applicable external requirements. Requirements vary by application and are not inferred from the corpus's abstract mathematics. This dossier does not attempt a medical, safety-critical or legal compliance assessment.

Evidence required: a domain-specific engineering review and integration tests. **Current status:** future work, not a property of the delivered software.

9.1 Three bounded experiments to prioritize

Experiment A — Recipe versus event table. Produce the same finite control sequence from a stored table and from the grammar/interpreter. Compare total storage and replay cost, including every dependency. This directly tests the compact-representation proposition.

Experiment B — Topology as an explicit option. Run otherwise identical state/control mappings with an ordinary periodic chart and the declared Klein quotient. Compare recurrence, seam behavior and output metrics. This tests whether the topological layer adds controllable structure instead of simply adding complexity.

Experiment C — Geometry adapter versus physical solver. Feed matched benign geometries and source schedules to an independent solver through a conventional interface and through the engine's adapter. Compare represented parameters, simulation outputs and setup cost. This tests integration without assuming the grammar has replaced wave physics.

These are proposed experiments, not completed trials. A favorable result in any one would support only its defined claim.

10. Attribution, positioning and unresolved decisions

10.1 A defensible positioning statement

Tom Klootwijk’s supplied corpus proposes an integrated, grammar-driven method for organizing multiscale geometric and time-dependent signal representations. Its defining conceptual move is to let a compact binary production system drive structured interpretation, rather than treating an expanded grid or sample array as the only primary representation. The corpus links that move to log-polar addressing, implicit boundaries, sound-cone geometry and non-orientable recirculation. [S1, pp. 19–22; S2, lines 48–99]

The present formalization turns that description into a research specification with typed interfaces and a reproducible subset implementation. The meaningful development proposition is the integration and usefulness of these components for specific tasks. No claim is made here that L-systems, signed-distance methods, log-polar coordinates, delta-sigma conversion, the golden ratio or the Klein bottle were newly invented in this corpus.

Originality or protectability of a particular combination would require a separate, evidence-based assessment. File hashes record the delivered bytes; they do not establish first conception, exclusive authorship, ownership, a filing date or legal priority. The supplied identifier is not treated as a verified registration number.

10.2 Decisions required for the next specification edition

The next edition should choose whether 1 rotates, branches or selects a profile-dependent operation; whether the log-polar texture remains a required backend or an optional cache; which numerical representation realizes the float-free ambition; what field is actually being measured by “SDF”; and whether the Klein construction is an indexing surface, an orientation-tagged state space or a separately realized physical structure.

It should also select one primary application and one falsifiable advantage. Without that selection, “broad-scale” remains a portfolio of possibilities. With it, the grammar, state representation and output adapter can be compared against a clear baseline.

10.3 Contribution record for this edition

The requester supplies the corpus, the requested attribution and the request for formalization. The original artifacts retain their exported dialogue structure. This edition adds the working definition, explicit type separation, formal model, derivations, claim register, application matrix, reference implementation, tests and packaging methodology with AI assistance.

The originals are preserved. Newly introduced conventions are labelled. No certificate of ownership, patent document, notarized declaration or independent scientific endorsement is implied by the presentation.

Development thesis

The strongest next claim is not that one-bit topology abolishes physical limits. It is that a carefully specified grammar and interpreter can provide a compact, controllable and reusable representation for a chosen class of geometric and signal tasks. That is a proposition the accompanying package makes concrete enough to investigate.

Appendix A. Claim register

The complete register is supplied in `data/claim_register.csv` and `.json`, with identifiers, source locators, status, treatment and validation requirements. The digest below is designed to keep the authorial statement and the engineering treatment adjacent.

C01 — Binary production rules

Source statement: $0 \rightarrow 1; 1 \rightarrow 01$. [S1 pp. 21-22; S2 lines 83-93]

Status: Specified; reproduced. **Treatment:** Use the exact substitution, not the reversed conventional Fibonacci word. The seed is not specified by the corpus.

Required check: Exact finite-generation fixtures and count recurrence.

C02 — Golden-ratio count limit

Source statement: The ratio of zeroes to ones converges to phi about 0.618. [S1 p. 22; S2 lines 89-93]

Status: Derivable with stated counts. **Treatment:** Use reciprocal golden ratio $\phi = (\sqrt{5}-1)/2$; distinguish $\Phi = 1/\phi$. This is a count ratio, not a general optimization result.

Required check: Check recurrence and large-depth integer counts.

C03 — One-bit data layer

Source statement: One-bit jitter / delta-sigma replaces heavy amplitude floats. [S1 p. 19; S2 lines 50-57]

Status: Partly specified; terminology unresolved. **Treatment:** Separate instruction bits, signal bits, quantizer state, amplitude dither, clock jitter and coordinate precision.

Required check: Specify clock, quantizer loop, filter, numeric formats and passband error.

C04 — One bit indexes a texture

Source statement: The stream directly indexes a two-dimensional log-polar LUT. [S1 pp. 7,19; S2 lines 54-57]

Status: Interface missing. **Treatment:** An isolated bit selects two alternatives; a stream needs a decoder/state to address a larger texture.

Required check: Publish address state, dimensions, channels, wrap and interpolation rules.

C05 — Log radius and octaves

Source statement: Equal radial distances and doubling the radius both describe octave spacing. [S1 p. 7; S2 line 57]

Status: Ambiguous conventions. **Treatment:** Retain both source phrases. Proposal separates linear radius R from logarithmic coordinate ρ ; octave is $\rho+1$ for $\rho = \log_2(f/f_{\text{ref}})$.

Required check: Round-trip tests; explicit units and coordinate transform.

C06 — Spacetime direct sum

Source statement: R^3 direct-sum R constitutes Minkowski spacetime. [S1 pp. 16-19; S2 lines 66-72]

Status: Requires additional structure. **Treatment:** Direct-sum decomposition supplies coordinates, not the metric; declare a quadratic form and distinguish acoustic analogy from relativistic spacetime.

Required check: Dimensional analysis and metric declaration.

C07 — SDF equals interval

Source statement: SDF 0, wavefront and null interval are identified. [S1 pp. 18-19; S2 lines 69-72]

Status: Shared zero set only in specified example. **Treatment:** q has squared-length units. $d = r - c_s(t-t_0)$ is a spatial-slice distance for future times; q and d are not identical fields.

Required check: Shared-zero and unequal-off-boundary numerical checks.

C08 — Geometry chain

Source statement: Apex, circle, pyramid, cone and sphere form a single projected structure. [S1 pp. 19-20; S2 lines 59-65]

Status: Conceptual; mappings incomplete. **Treatment:** Separate constant-time slices, dimensional projections, causal cones and faceted approximations. A circle swept in one time direction alone gives 2+1 dimensions.

Required check: Supply each domain/codomain, metric and approximation tolerance.

C09 — Blend formula

Source statement: $\text{Blend} = \min(a,b) - \max(k - |a-b|, 0)^2 / (4k)$. [S1 p. 14; S2 lines 79-81]

Status: Specified; reproduced. **Treatment:** Formula is polynomial, not exponential. $k > 0$; all operands share units. Smooth blending changes the original zero set.

Required check: Symmetry, support, $k > 0$ and $\text{Blend}(0,0) = -k/4$.

C10 — Interchangeable operators

Source statement: Addition, smin , direct sum and XOR describe the same blend. [S1 pp. 13-17,20; S2 lines 66-81]

Status: Different semantic types. **Treatment:** Use superposition for linear pressure signals, min/smin for geometric fields, direct sum for spaces, XOR for bits.

Required check: Type-check interfaces; do not substitute operators across types.

C11 — Global Klein-bottle SDF

Source statement: A globally inside/outside signed field is attached to a Klein bottle. [S1 pp. 20-22; S2 lines 74-81]

Status: Requires replacement representation. **Treatment:** Proposal uses a quotient chart plus orientation parity. This is not a globally sided embedded Klein-bottle SDF in ordinary three-space.

Required check: Seam identity, chart parity and scalar-consistency tests.

C12 — Topological phase inversion

Source statement: Crossing the surface automatically flips acoustic phase. [S1 pp. 20-22; S2 lines 78,93]

Status: Not implied. **Treatment:** Orientation reversal is a chart property. A scalar pressure has no compulsory sign reversal; a twisted state needs an explicit coupling rule.

Required check: Compare scalar and orientation-tagged transports.

C13 — Recirculation means energy retention

Source statement: The topology produces a perfectly stable self-contained energy engine. [S1 p. 20; S2 lines 76-81]

Status: Physical claim unverified. **Treatment:** Recurrence of computational coordinates is not a physical energy balance or stability proof.

Required check: Define input, storage, losses, radiation and feedback; measure decay/stability.

C14 — Grammar determines metric distance

Source statement: Word depth and length alone supply distance to SDF 0. [S1 p. 22; S2 line 93]

Status: Insufficient data. **Treatment:** A word is an address only relative to a decoder, metric, geometry and initial state. Many different points share word length/depth.

Required check: Counterexample with equal-length paths; validate a declared distance oracle.

C15 — Zero aliasing / infinite resolution

Source statement: No grid or float implies zero aliasing and infinite resolution. [S1 pp. 5-8,22; S2 lines 95-99]

Status: Not a demonstrated guarantee. **Treatment:** Finite output clocks, finite numeric states and nonlinear operations still require error and bandwidth control.

Required check: Spectrum, reconstruction error, numeric error and sample-rate convergence.

C16 — Near-zero memory

Source statement: Binary grammar removes nearly all memory cost. [S1 pp. 21-22; S2 lines 153-156]

Status: Conditional compression only. **Treatment:** A short recipe can replace expanded data; runtime state, emitted output, decoder, caches and buffers remain.

Required check: Report total memory separately from packed grammar bytes.

C17 — Golden ratio eliminates bad resonances

Source statement: Golden-ratio splits prevent unwanted destructive resonances. [S1 p. 20; S2 lines 24,149]

Status: Unverified performance claim. **Treatment:** Count ratios and geometric partitions do not establish transfer-function behavior or universal optimality.

Required check: Controlled spectral and application-specific comparison with baselines.

C18 — Caustic solution

Source statement: An SDF gradient and named normal caustic coordinates give exact focused amplitude. [S1 pp. 10-11]

Status: Not specified as a solver. **Treatment:** Corpus supplies no amplitude/transport solver, coefficient data or reproducible definition of NCCS. Distance alone is insufficient.

Required check: Specify wave model and compare finite-frequency solution near focus.

C19 — Direct grammar-to-power output

Source statement: Grammar bits directly drive switching hardware to obtain a pure physical wave. [S1 p. 22; S2 lines 95-99]

Status: Physical interface missing. **Treatment:** Instruction code and modulation code are different. Proposal includes only an offline quantizer, no power-stage interface.

Required check: Verify modulation, reconstruction and hardware independently before deployment.

C20 — Beam invariance / physical limitations

Source statement: Representation bypasses aperture and propagation limits. [S1 pp. 8-10; S2 lines 137-149]

Status: Unverified; not operationalized. **Treatment:** Geometry representation is not evidence of frequency-independent beam performance or hardware protection. Hazardous uses remain archival.

Required check: Benign simulation and professionally controlled measurement only.

C21 — Tree replacement

Source statement: The phi-parameterized search tree is replaced by grammar execution. [S1 pp. 20-22; S2 lines 83-93]

Status: Explicit source progression. **Treatment:** Preserve the tree as a historical stage, not a mandatory final runtime component. Optional indexing would be a new engineering extension.

Required check: Architecture traceability inspection.

C22 — Branch or rotate

Source statement: The symbol 1 means branch or golden-ratio rotation. [S1 p. 21; S2 lines 87-89]

Status: Underspecified operational semantics. **Treatment:** Reference profile chooses rotation only and declares the angle. Branching requires state-fork/return and budget rules.

Required check: Interpreter fixture tests and explicit profile documentation.

Appendix B. Source map and primary references

B.1 Corpus locator map

Corpus region	Main content	Dossier use
S1, pp. 1–2	General tone/search preface	Context, not engine evidence
S1, pp. 2–4	Note, frequency, keys and phase accumulation	Data-layer history
S1, pp. 5–8	Procedural fields and log-polar texture	Coordinate/field intent and ambiguity
S1, pp. 8–11	Directional acoustics and caustics	Source claims; non-operational treatment
S1, pp. 12–16	Zero, metaphors, blending, direct sum, XOR	Lexicon and operator distinction
S1, pp. 17–19	Space/time and sound-cone construction	Declared acoustic analogue
S1, pp. 19–20	One-bit data, geometry, Klein bottle, tree	Pre-grammar architecture
S1, pp. 21–22	Binary L-system and tree replacement	Final source-stage priority
S1, p. 23	Export tail	No substantive engine specification
S2, lines 48–99	Five-layer formalized restatement	Organization of Section 2
S2, lines 131–133	Meaning of corpus	Scope clarification
S2, lines 135–162	Application discussion	Portfolio basis, with limits stated

S1 and **S2** are the user-supplied primary corpus artifacts, not independent technical validation. Their SHA-256 values and byte sizes are in `sources/source_index.json`. Original diagrams and equations were inspected in the supplied page renderings; no OCR was used.

B.2 External primary references

All external sources were consulted on 10 September 2026. These references clarify components and comparison methods; none reports tests of Tom Klootwijk's integrated engine. Bibliographic data and links are also in `sources/references.json` and `sources/references.bib`.

R1. Przemysław Prusinkiewicz and Aristid Lindenmayer. *The Algorithmic Beauty of Plants, Chapter 1: Graphical modeling using L-systems*. 1990. [Sections 1.2-1.3; rules and geometric interpretation are separate.] [Primary-source access](#).

R2. Analog Devices. *MT-022: ADC Architectures III, Sigma-Delta ADC Basics*. n.d.. [Oversampling, noise shaping, filtering and decimation; Figure 3, PDF p. 5.] [Primary-source access](#).

R3. Analog Devices. *MT-004: The Good, the Bad, and the Ugly Aspects of ADC Input Noise*. n.d.. [Dither is an added signal/noise process with conditional effects.] [Primary-source access](#).

R4. Analog Devices. *MT-007: Aperture Time, Aperture Jitter, Aperture Delay Time: Removing the Confusion*. n.d.. [Timing uncertainty is distinct from amplitude dither.] [Primary-source access](#).

R5. David Tong. *Dynamics and Relativity, Chapter 7: Special Relativity*. n.d.. [Metric structure and Minkowski space; PDF p. 20.] [Primary-source access](#).

R6. John C. Hart. *Sphere Tracing: A Geometric Method for the Antialiased Ray Tracing of Implicit Surfaces*. author manuscript; journal DOI 10.1007/s003710050084. [Section 2.1: geometric distance, bounds and Lipschitz conditions.] [Primary-source access](#).

R7. Jean Gallier and Dianna Xu. *A Guide to the Classification Theorem for Compact Surfaces*. 2013. [Section 1.2; nonorientable surfaces and Klein-bottle edge identifications.] [Primary-source access](#).

R8. k-Wave project; Bradley Treeby, Ben Cox and collaborators. *k-Wave documentation: overview and physical model*. accessed 2026-09-10. [Independent acoustic solver with explicit equations and material parameters.] [Primary-source access](#).

R9. Antonio Stanziola, Simon R. Arridge, Ben T. Cox and Bradley E. Treeby. *j-Wave: An open-source differentiable wave simulator*. 2022. [Modular differentiable acoustic simulation; precedent for simulator adapters, not validation of this corpus.] [Primary-source access](#).

External research did not resolve the corpus-specific NCCS construction or supply physical validation of the complete architecture. The terminology is consequently preserved as unresolved rather than replaced with a confidently asserted external equivalent.

Appendix C. Package guide and conformance profile

C.1 Delivered directory structure

```

klootwijk_corpus_package/
  README.md
  metadata.json
  LICENSE_NOTE.md
  SHA256SUMS.json
  report/      PDF, editable Markdown, LaTeX, design inputs
  engine/      standard-library Python demonstrator
  tests/       35 automated tests
  examples/    generated files, result record, test log
  data/        claim register, application matrix, profile
  sources/     original uploads, derived text, references
  tools/       PDF build and package verification scripts

```

C.2 Reference profile P0

Parameter	Value or convention	Provenance
Productions	$0 \rightarrow 1, 1 \rightarrow 01$	Source
Initial word	0	Proposal
Demo depth	12	Proposal
Interpretation	0 moves/ticks; 1 rotates, no branch	Source-inspired proposal
Spatial step / grammar-time step	0.125 / 0.01	Proposal; dimensionless chart / grammar time
Turn angle	$2\pi(1 - \phi)$	Proposal
Chart	$(u + 1, v) \sim (u, -v)$ and $(u, v + 1) \sim (u, v)$	Declared quotient
Pitch reference	440 Hz	Chosen example
Sample clock	16,000 Hz; 16,000 samples	Chosen example
Arithmetic	Python floating point except exact grammar/counts	Explicit implementation choice
Bit packing	MSB first; zero-pad last byte; store logical count	Proposal
Output access	Files only; no playback or device access	Implementation boundary

C.3 Rebuild and integrity

The report can be rebuilt with `bash tools/build_report.sh` when Pandoc, XeLaTeX and the named system fonts are installed. Font files are not distributed. The editable Markdown is the content source; `report/corpus_formalization.tex` is its generated typesetting representation.

Run `python tools/verify_package.py` before editing or regenerating examples. The manifest covers the delivered files except the manifest itself. Hash verification checks byte integrity against that manifest, not authenticity, date of original creation or ownership. The archive has no passwords, network callbacks or executable installation steps.

Privacy: the PDF and metadata retain the name, identifier and date supplied in the request. The original uploads are also included. Review all of these before publishing the archive.

End of edition 1.0.